



Stop training, start prompting

The future of dynamic video surveillance



zenika



BACKGROUND

The security sector faces CCTV monitoring challenges

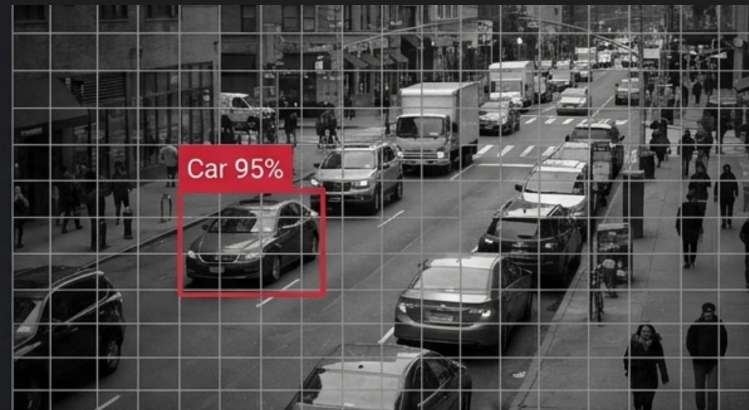
- Managing growing CCTV networks with limited manpower is difficult and costly
- Operators struggle with isolated and siloed video feeds
- Searching across multiple videos for timely insights is challenging





Traditional Computer Vision

Bounding boxes and confidence scores



1-See it once

Scans the video frame in a single pass

2- Draw the box

Place a bounding box around the objects to be detected

3-Score the match

Assign a confidence level to the match

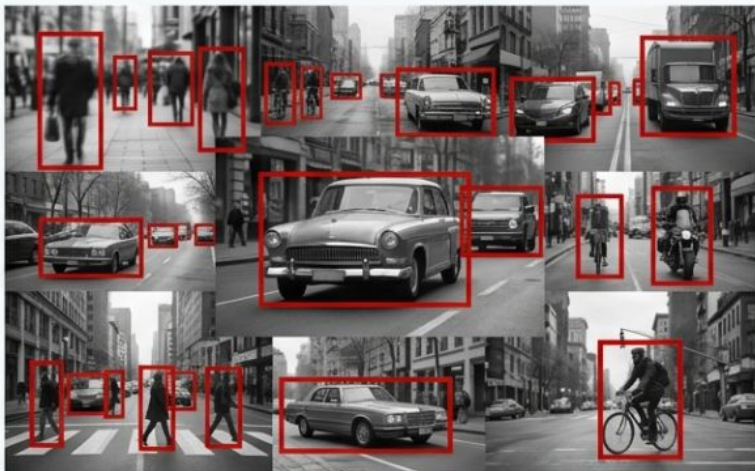
4-Trigger action

Send an alert and other programmed actions

Fixed Dictionary

Relies on predefined categories

The Dataset



Millions of images manually gathered.
Painstakingly boxed and labeled.

The Output

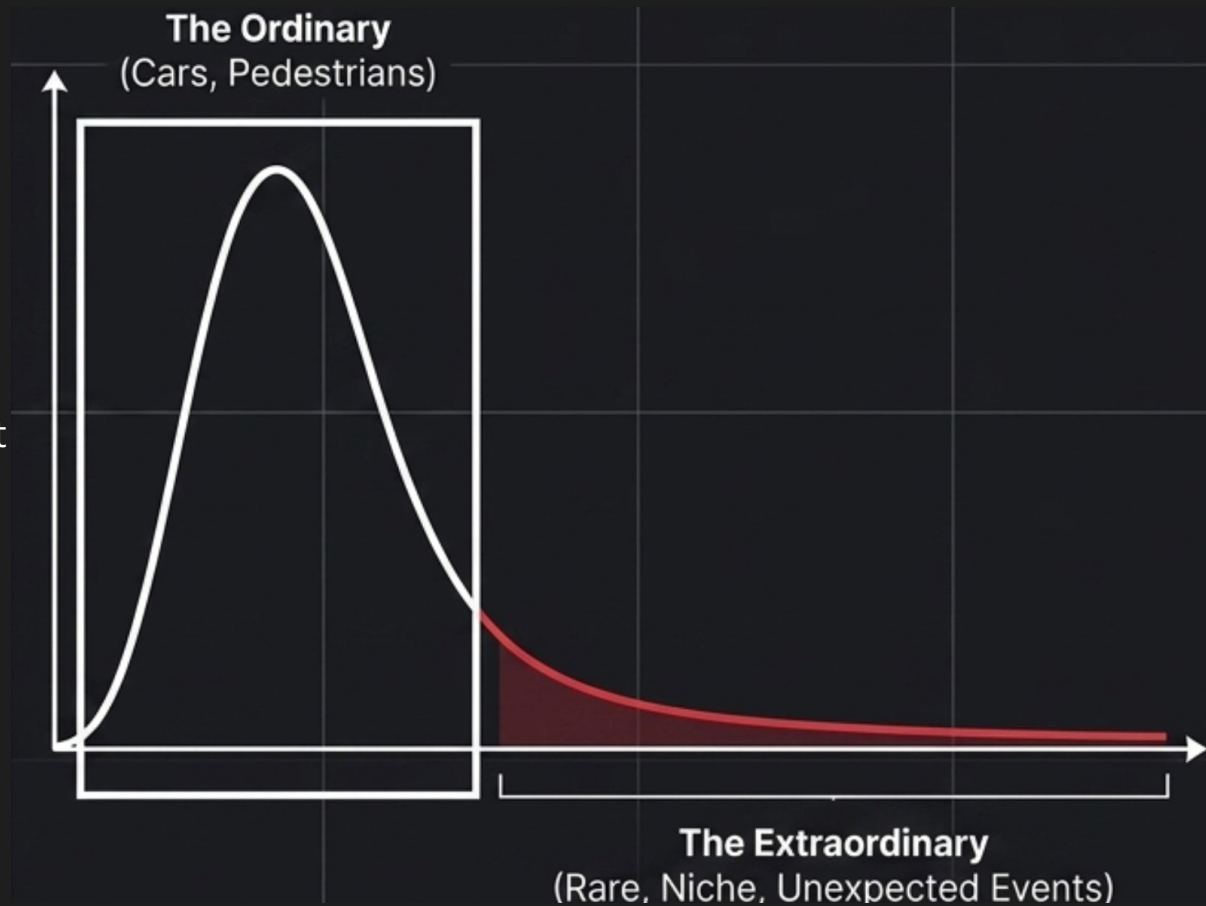


Models are locked to a fixed list.
Blind to anything outside that exact menu.

The Unexpected

Long-tailed problem

- Model confident on trained material
- Cannot proactively collect training data for all anomalies
- Many security incidents are rare events

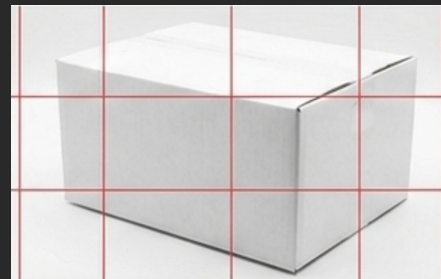
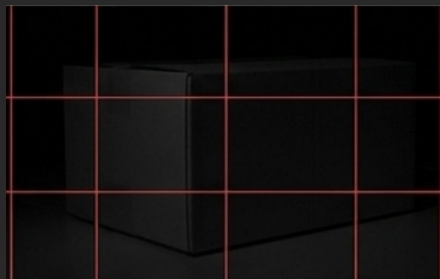
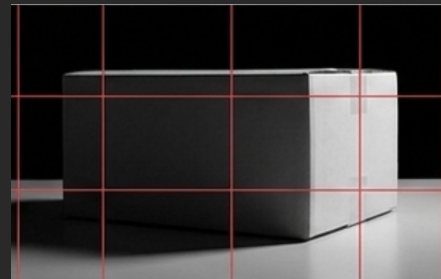
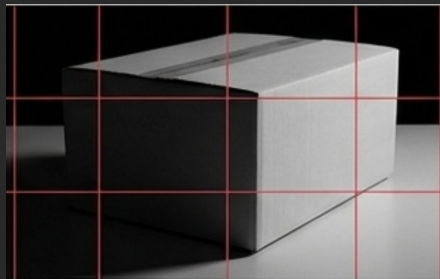
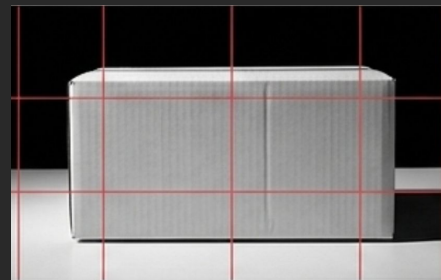
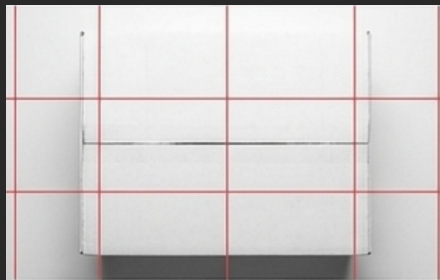




Retraining Costs

Exhaustive data required

- Data hunger
- Variation is key to success
- Burden of data curation
- Process takes time



Compute Costs

Expensive iterations

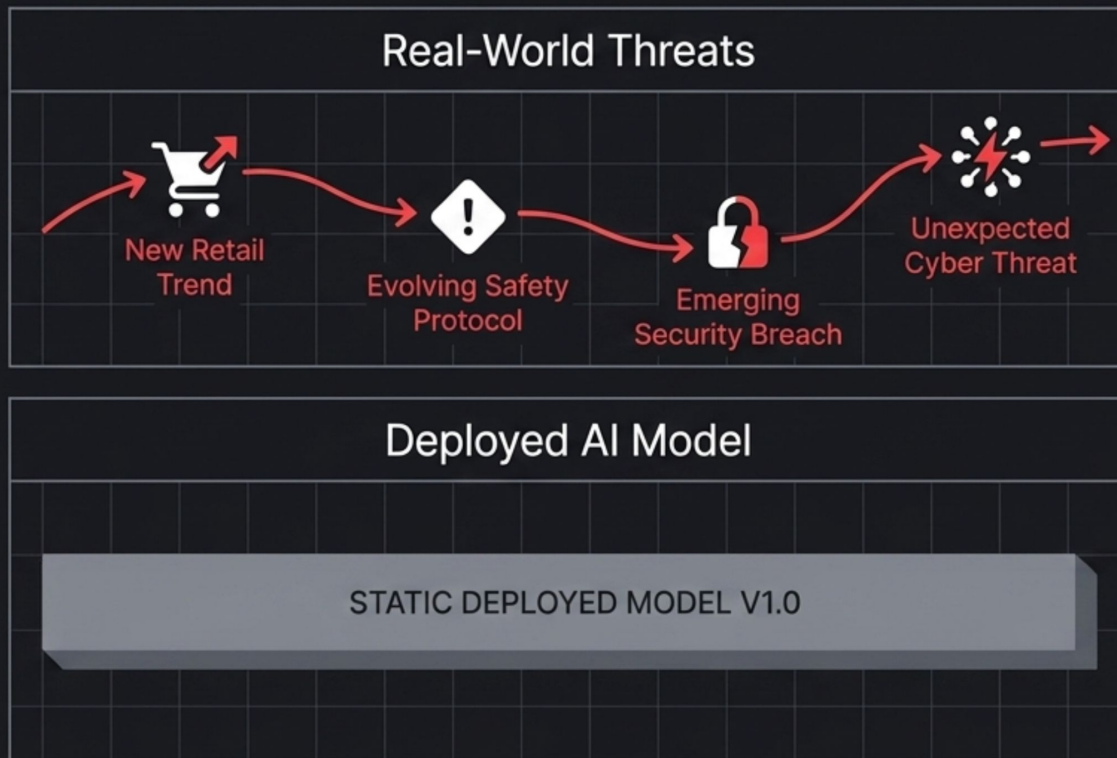
Business may need to wait up to weeks to deploy a model update that was needed yesterday



Static models

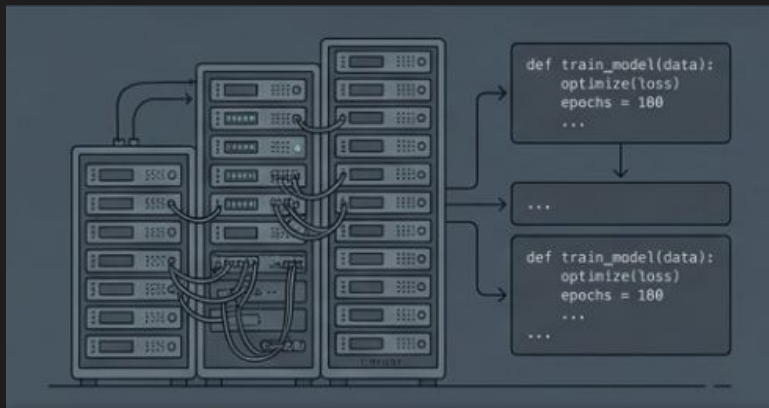
Challenges in a dynamic world

- The world changes
- The model is frozen
- Reactive



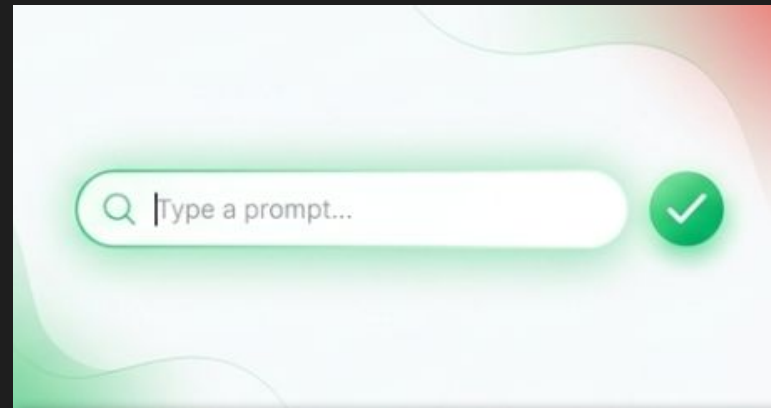
The Paradigm Shift

Transition from training to prompting



Training Models

- Write code
- Curate datasets
- Train models
- Wait weeks for deployment

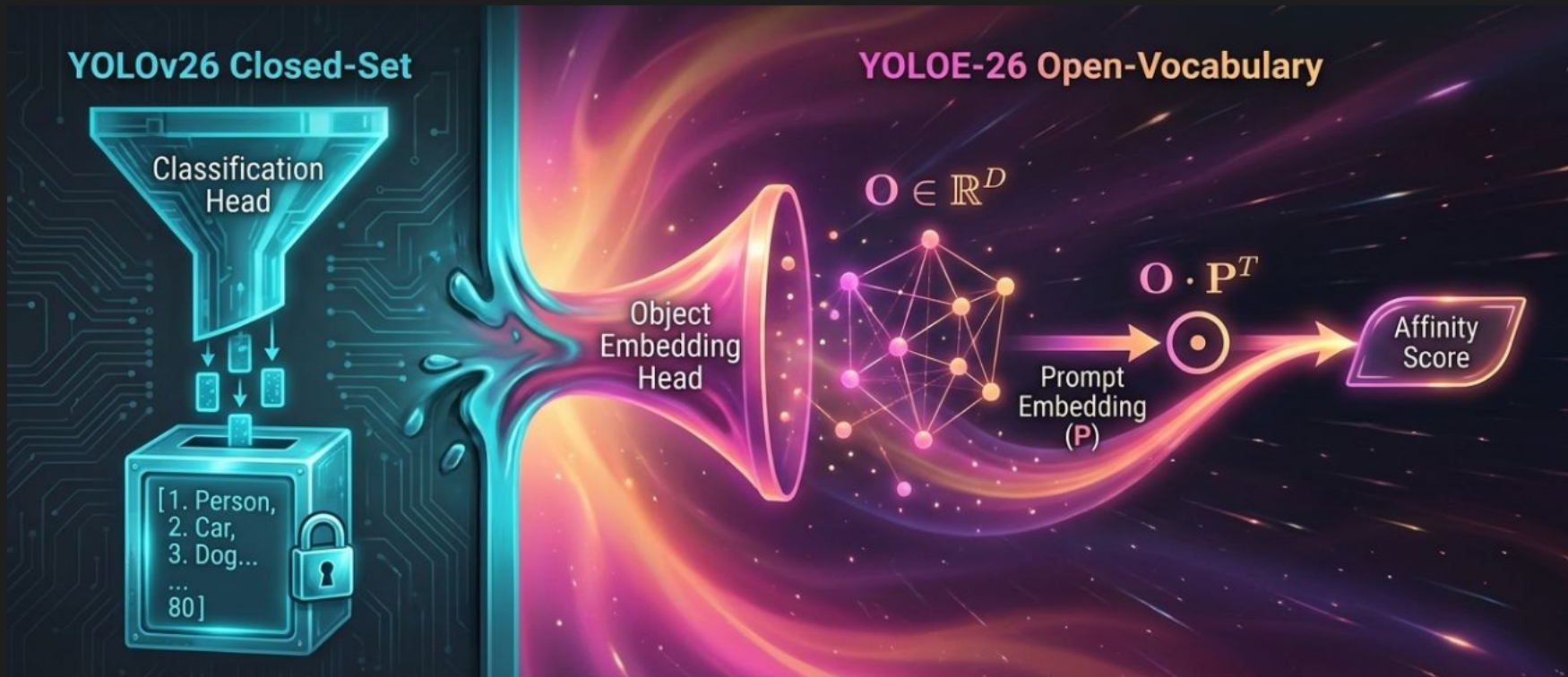


Prompt Models

- Type a prompt
- Instantly search
- Real-time user application

Enter YOLOE-26

Open vocabulary unified embedding space



Use Cases: Zero-Shot Prompting

Deduces what you are looking for without prior training

> "yellow suitcase"



Semantic Prompting

> "persons fighting"



Abstract Concepts

Visual Prompting

Open vocabulary unified embedding space



Visual Signature
Fingerprint



 **No Typing Required:**
Sometimes an image is worth a thousand words.

 **Select the Target:**
Click directly on an object or person of interest.

 **Visual Signature:**
Generate a unique fingerprint for that specific target.

 **Seamless Tracking:**
The AI automatically isolates and tracks the target.

Cross-Camera Search

Cameras no longer operate as isolated feeds



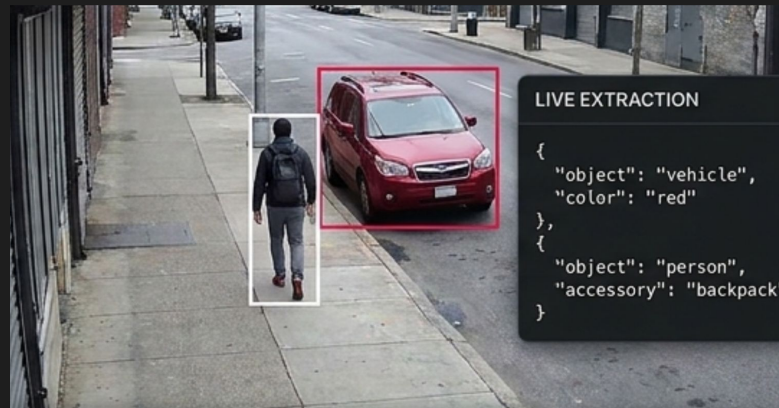


Summary



The problem

- Unprecedented volumes of unstructured video data
- Challenging to extract timely insights across multiple videos.



Computer vision tools

- AI object detection models
- Semantic text search prompts
- Visual prompting across multiple videos

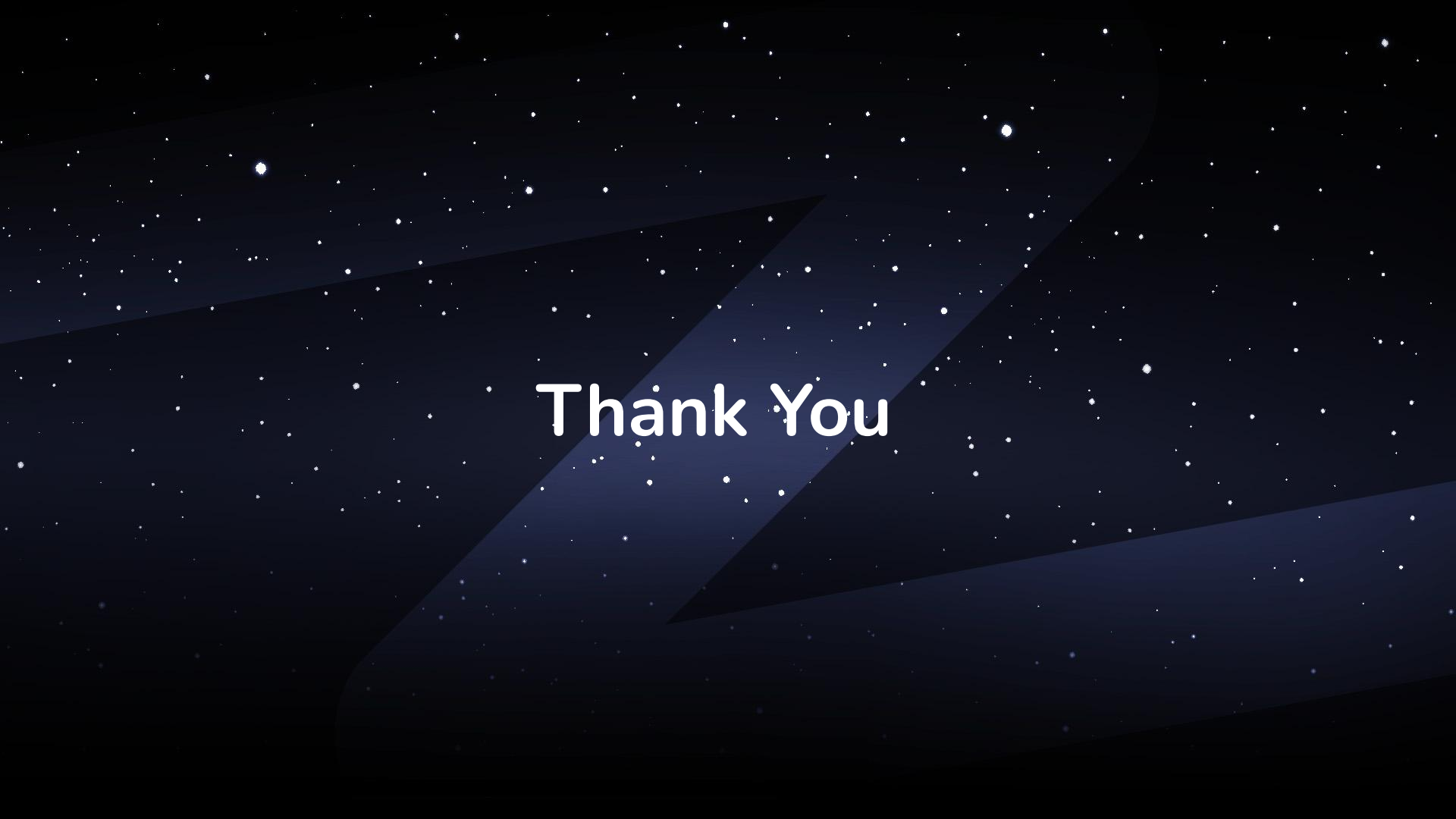
Q & A



LinkedIn



Presentation & Source

The background is a deep navy blue, speckled with numerous small, bright white dots representing stars. Overlaid on this are several large, semi-transparent, dark blue geometric shapes that create a sense of depth and movement. These shapes include a large, curved band sweeping from the top left towards the center, and another band below it, creating a layered effect. The text "Thank You" is centered in a clean, white, sans-serif font.

Thank You